

$$\equiv p \vee (\neg p \vee q) \text{ (By 1st commutative law)}$$

$$\equiv (p \vee \neg p) \vee q \text{ (By 1st associative law)}$$

$$\equiv T \vee q \text{ (1st negation law)}$$

$$\equiv q \vee T \text{ (1st commutative law)}$$

$$\equiv T \text{ (1st Domination Law)}$$

$$b) p \rightarrow (p \vee q) \equiv \neg p \vee (p \vee q) \text{ (By conditional-disjunction equivalence)}$$

$$\equiv (\neg p \vee p) \vee q \text{ (1st associative law)}$$

$$\equiv (p \vee \neg p) \vee q \text{ (1st commutative law)}$$

$$\equiv T \vee q \text{ (1st negation law)} \equiv q \vee T \text{ (1st commutative law)} \equiv T \text{ (1st Domination Law)}$$

$$c) \neg p \rightarrow (p \rightarrow q) \equiv \neg(\neg p) \vee (p \rightarrow q) \text{ (By conditional-disjunction equivalence)}$$

$$\equiv p \vee (p \rightarrow q) \text{ (By Double-negation law)}$$

$$\equiv p \vee (\neg p \vee q) \text{ (By conditional-disjunction equivalence)}$$

$$\equiv (p \vee \neg p) \vee q \text{ (By 1st associative law)} \equiv T \vee q \text{ (1st negation law)}$$

$$\equiv q \vee T \text{ (1st commutative law)} \equiv T \text{ (1st Domination Law)}$$

$$d) (p \wedge q) \rightarrow (p \rightarrow q) \equiv \neg(p \wedge q) \vee (p \rightarrow q) \text{ (By conditional-disjunction equivalence)}$$

$$\equiv \neg(p \wedge q) \vee (\neg p \vee q) \text{ (By conditional-disjunction equivalence)}$$

$$\equiv (\neg p \vee \neg q) \vee (\neg p \vee q) \text{ (1st De Morgan's Law)}$$

$$\equiv ((\neg p \vee \neg q) \vee \neg p) \vee q \text{ (1st Associative Law)}$$

$$\equiv (\neg p \vee (\neg p \vee \neg q)) \vee q \text{ (1st Commutative Law)}$$

$$\equiv ((\neg p \vee \neg p) \vee \neg q) \vee q \text{ (1st Associative Law)}$$

$$\equiv (\neg p \vee \neg q) \vee q \text{ (1st Idempotent Law)}$$

$$\equiv \neg p \vee (\neg q \vee q) \text{ (1st Associative Law)}$$

$$\equiv \neg p \vee (q \vee \neg q) \text{ (1st Commutative Law)}$$

$$\equiv \neg p \vee T \text{ (1st Negation Law)}$$

$$\equiv T \text{ (1st Domination Law)}$$

$$e) \neg(p \rightarrow q) \rightarrow p \equiv \neg(\neg(p \rightarrow q)) \vee p \text{ (By conditional-disjunction equivalence)}$$

$$\equiv (p \rightarrow q) \vee p \text{ (By Double-negation Law)} \equiv (p \vee q) \vee p \text{ (By conditional-disjunction equiv)}$$

$$\equiv p \vee (\neg p \vee q) \text{ (1st commutative Law)} \equiv (p \vee \neg p) \vee q \text{ (1st associative Law)}$$

$$\equiv T \vee q \text{ (1st Negation Law)} \equiv q \vee T \text{ (1st Commutative Law)} \equiv T \text{ (1st Domination Law)}$$